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Speedup signing: pre-rejection sampling towards dilithium

Lianglin Yan^{1,2*} , Ming Luo^{1,2} and Mingsheng Wang¹

Abstract

Security and efficiency have always been two critical factors in the development of post-quantum digital signatures. As the best-known scheme, Dilithium (Ducas et al., TCHES 2018) is SUF-CMA in QROM and has a relatively fast efficiency with many untrivial optimizations. The goal of this paper is to propose some techniques that can promote signing speed without sacrificing security. We first propose the pre-rejection sampling technique in KeyGen stage to reduce the rejections of the fourth condition, consequently resulting in some speedup in Sign stage. To prove security, we propose the c -selected MLWE problem, a variant of MLWE that can offer the equivalent security as original MLWE. Applying these two techniques to Dilithium, we obtain an advanced signature scheme with better efficiency, and without any other losses except some pre-computations. Security reduction demonstrates that our scheme is also SUF-CMA in QROM. The experimental results show that pre-rejection sampling achieves a 47%, 22%, and 17% reduction in the rejection probability of the fourth condition over Dilithium scheme when the parameter set corresponds to NIST's security levels 2, 3 and 5, respectively. This type of reduction increases signing speed by approximately 1% under the parameter set 2 of Dilithium.

Keywords Post-quantum signature, MLWE, Pre-rejection sampling

Introduction

Post-quantum cryptography (PQC) has gradually become a research hotspot in recent decades, whose goal is to defend against adversaries with quantum attack capabilities. According to the result of the third round of NIST's post-quantum cryptography algorithms competition (2024), lattice-based schemes are seemly most promising, and this paper is exactly focusing on latticed-based post-quantum signatures.

At first, several kinds of signature schemes based on lattice assumptions are proven to be secure and practical in (classical) random oracle model (ROM). Parts of them use trapdoor functions and hash-and-sign method

(Gentry et al. 2008; Stehlé and Steinfeld 2011) while the others obtain signatures by combining the canonical authentication protocols and Fiat-Shamir transformations (Lyubashevsky 2009, 2012; Güneysu et al. 2012; Bai and Galbraith 2014; Ducas et al. 2018), and the latter is more popular. The Fiat-Shamir paradigm was first proposed in Fiat and Shamir (1987), and further applied to authentication protocols by Abdalla et al. Abdalla et al. (2002). A series of improved works have been carried out by modifying signature schemes based on a variety of different hard problems on lattices, i.e., Short Integer Solution (SIS) problem, Learning with Errors (LWE) problem, and other analogs. More precisely, Lyubashevsky implemented the "Fiat-Shamir with aborts" transformation based on the Ring-SIS (RSIS) problem (Lyubashevsky 2009) and provided an alternative signature scheme based on LWE and SIS problems. He also developed the rejection sampling technique for higher efficiency. Güneysu et al. Güneysu et al. (2012) and Bai and Galbraith Bai and Galbraith (2014) are both dedicated to

*Correspondence:

Lianglin Yan
yanlianglin@iie.ac.cn

¹ Key Laboratory of Cyberspace Security Defense, Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China

² School of Cyber Security, University of Chinese Academy of Sciences, Beijing, China

exploring signature compression methods. However, none of these signatures are provably secure in a quantum environment.

Post-quantum Fiat-Shamir transformations have been studied in many works (Boneh et al. 2011; Boneh and Zhandry 2013; Unruh 2017; Kiltz et al. 2018). They try to construct reductions from underlying intractable problem to signature schemes in the quantum random oracle model (QROM), but there is still a gap remained until Liu and Zhandry's proof (Liu and Zhandry 2019), whose results show that former Fiat-Shamir signatures are quantumly secure without any modifications. Consequently, CRYSTALS-Dilithium (Bai et al. 2021) is totally accepted by all because of its most efficient construction. Besides security, Dilithium also contains many considerations regarding the efficiency of transmission and implementation. We ask:

Is there any room for improvement in Dilithium scheme under such a fixed signature pattern?

We target a higher efficiency than the state of the art.

The most time-consuming part

The rejection sampling technique was originally proposed for security and efficiency, but it gradually became an obstacle with the improvement of signature scheme. In order to generate a valid signature, Dilithium should run *Sign* algorithm around four times on average. Signer will discard the signature values and intermediate variables that do not meet the conditions and re-run the whole *Sign* program. To our knowledge, there is no effective way to reduce the number of rejections in the sign stage except by adjusting the parameters. Moreover, adjusting parameters is just a trade-off between security and efficiency, i.e., a higher probability of passing rejection conditions always indicates lower security.

Our contributions

In this work, we propose two new techniques to promote Dilithium and obtain a Fiat-Shamir signature scheme with better efficiency and no security loss. We accelerate the signing speed by introducing pre-rejection sampling in *KeyGen* procedure, which lowers the loops of rejection sampling steps of *Sign* stage. We also introduce the c -selected MLWE assumption to ensure the security of our scheme. Security reduction shows that our signature scheme is SUF-CMA (Strong UnForgeability under Chosen Message Attacks) in QROM. As for performance, it can achieve a faster signing speed than Dilithium due to fewer rejections in *Sign* procedure. What's more, all technical details and optimizations of the Dilithium are compatible with our improvements, which means a high portability.

Pre-rejection sampling for higher *sign* speed

There are two layers of rejection conditions when Dilithium produces signature values. To reduce the number of rejection sampling and speed up *Sign* procedure, we develop a pre-rejection technique that acts on the key generation stage. For rejection conditions $\|ct_0\| \geq \gamma_2$ and “# of 1's in h is greater than ω ”, we conduct a precise analysis showing how much impact these two conditions really have on the overall rejection sampling. Furthermore, we notice that the passing probability increases when t_0 is selected with some preset conditions during the *KeyGen* procedure. Specifically, by forcing the coefficients of t_0 to follow a unimodal distribution with a relatively small variance σ^2 , we achieve a much smaller variance of ct_0 's coefficients and consequently, fewer 1's in h . In some cases, pre-rejection sampling halves the rejection probability of the inner round conditions, i.e., the third and fourth conditions.

New MLWE variant

Dilithium uses Hermite Normal Form (HNF) of MLWE problem to ensure that secret key recovery is impossible. However, our pre-rejection sampling decreases the possible set of public key, which means the output of *KeyGen* is not MLWE instance anymore. To solve this problem, we propose the c -selected MLWE problem to characterize the new public key. It employs a c -select function based on original MLWE. Hardness reduction implies that such c -selected MLWE is as hard as the HNF of MLWE.

An improved signature scheme

Employing pre-rejection sampling technique, we obtain a Fiat-Shamir signature with reduced number of rejections in *Sign*, thereby speeding up the signing. The correctness proof of the new signature scheme is almost the same as that of Dilithium since our modification is just a filtration. Note that our scheme is fully compatible with Dilithium, one can transfer a Dilithium system to ours by modifying *KeyGen* while remaining all other deployments, which provides a strong practicality.

Security proof and implementation

The security of our signature scheme relies on c -selected MLWE and MSIS problems. We prove it is SUF-CMA in QROM and is as secure as Dilithium when parameterized similarly. We provide three parameter sets corresponding to NIST's different security levels and conduct a comparison between Dilithium and ours. The experimental results show that our signing is slightly faster as we reduce the rejection probability (therefore the number of rejections). Pre-rejection sampling increases the running time of

KeyGen. However, in most real systems, one key is used for multiple signatures, allowing the increased KeyGen time to be amortized over numerous uses.

Development of dilithium

In this paper, we mainly focus on the Dilithium signature scheme. As a finalist in the third round of NIST's post-quantum cryptography algorithms competition (2024), Dilithium has been updated and evolved since its original publication. We introduce several important versions that help readers to know the state of the art. A non-original but relatively mature published version of Dilithium algorithm is (Ducas et al. 2018), it is contemporaneous with the submission to the first round of the NIST call for post-quantum standards (this can be viewed as Version 1). The Version 2 and Version 3 are respectively the submissions to the second and third rounds of the NIST competition, and the latest version is Version 3.1 Bai et al. (2021). Changes in these versions mainly involved the parameter selection and some details of implementation and security while the main steps and security assumptions of the signature algorithm are unchanged. The most up-to-date code implementation (Dilithium 2022) is about the Version 3.1, so we develop our work based on this version.

More recently, NIST has released the Module-Lattice-Based Digital Signature Standard (2024). The standard algorithm ML-DSA is derived from Version 3.1 (Bai et al. 2021) of Dilithium with some modifications related to the considerations about implementation details and practical security. The steps in the signature algorithm are consistent with those of Dilithium. So our proposed techniques that adjust the key generation step can also be applied to the ML-DSA algorithm and obtain the similar improvements.

As for security proof, all the versions of Dilithium (including the ML-DSA algorithm) have the same theoretical security, relying on the MLWE and MSIS assumptions. But all the versions of documentations mentioned above only contained a general sketch of security proof, the full version of security reduction was provided in Kiltz et al. (2018). Therefore, we will discuss the security of our proposed scheme based on Kiltz et al. (2018).

Organizations

Section 2 provides some preliminaries. In section 3, we present pre-rejection sampling technique and our signature scheme, also with the correctness proof. Section 4 contains the definition and hardness reduction of the c -selected MLWE assumption, and the security analysis of our scheme. Parameter selection and implementations are given in section 5. We also provide some other

related works in section 6 for the current state of the art. Finally, we conclude our work and discuss future work in section 7.

Preliminaries

Notations and distributions

In this paper, we let q be a prime integer, n be a power of 2. $\mathbb{Z}_q = \mathbb{Z} \cap (-q/2, q/2]$ denotes the integers modulo q . The underlying algebraic structure in this paper is cyclotomic ring $\mathcal{R} = \mathbb{Z}[x]/(x^n + 1)$ and $\mathcal{R}_q = \mathcal{R}/(q)$. We also enforce $q \equiv 1 \pmod{2n}$ so that $\mathcal{R}_q$ is NTT-friendly. We write lower-case letters as elements in $\mathcal{R}$ or $\mathcal{R}_q$, while bold-faced lower-case letters represent (column) vectors, e.g., $\mathbf{v} = (v_0, v_1, \dots, v_{m-1})^T$ where $v_i \in \mathbb{Z}$ or $\mathcal{R}$ and $\mathbf{v}^T$ is the transpose vector of $\mathbf{v}$. Naturally, for polynomial $a = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1} \in \mathcal{R}$, $\mathbf{a}$ denotes the column vector of coefficients of a , i.e., $\mathbf{a} = (a_0, a_1, \dots, a_{n-1})^T$. Sometimes without causing ambiguity, we also let $\mathbf{a}$ replace $\mathbf{a}$. Matrices over integers or polynomials are represented in bold-faced upper-case letters such as $\mathbf{M}$. The inner product of $\mathbf{a} = (a_0, a_1, \dots, a_{n-1})$ and $\mathbf{b} = (b_0, b_1, \dots, b_{n-1})$ is defined as $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{i=0}^{n-1} a_i \cdot b_i$.

Norms

The infinity norm (also called ℓ_∞ -norm) of m -dimensional vector $\mathbf{v}$ over integers is $\|\mathbf{v}\|_\infty = \max_{0 \leq i < m} |v_i|$. For an element $a \in \mathcal{R}$, $\|a\|_\infty$ denotes the infinity norm of the coefficients vector of the polynomial. For a vector $\mathbf{w} = (w_0, w_1, \dots, w_{m-1}) \in \mathcal{R}^m$, we define $\|\mathbf{w}\|_\infty = \max_{0 \leq i < m} \|w_i\|_\infty$. Other norms are defined similarly.

Modular reduction

For an even (resp. odd) positive integer α , we define centered reduction of r modulo α , denoted by $r' = r \bmod \pm\alpha$, to be the unique element in $(-\frac{\alpha}{2}, \frac{\alpha}{2}]$ (resp. $[-\frac{\alpha-1}{2}, \frac{\alpha-1}{2}]$) such that $r' \equiv r \pmod{\alpha}$. Similarly, $r' = r \bmod +\alpha$ denotes the the element in $[0, \alpha)$ such that $r' \equiv r \pmod{\alpha}$.

Distributions

We introduce the notation of some distributions from Bai et al. (2021). Let B_τ denote the set of elements in $\mathcal{R}$ that consist of τ coefficients which are either -1 or 1 , with all remaining coefficients being 0. For an integer η , $\mathcal{S}_\eta$ is the set which contains all the elements $w \in \mathcal{R}$ with ℓ_∞ -norm less or equal to η , i.e., $\mathcal{S}_\eta = \{w \in \mathcal{R} \mid \|w\|_\infty \leq \eta\}$. We define $\mathcal{U}(\mathcal{S}_\eta)$ to denote the uniform distribution over $\mathcal{S}_\eta$, and notation $s \leftarrow \mathcal{U}(\mathcal{S}_\eta)$ means sampling a polynomial by uniformly drawing its coefficients from the interval

$[-\eta, \eta]$. In addition, we define uniform distribution $\mathcal{U}(\tilde{\mathcal{S}}_\eta)$ similarly where $\tilde{\mathcal{S}}_\eta = \{w \bmod \pm 2\eta \mid w \in \mathcal{R}\}$.

Learning with errors

Regev Regev (2005) firstly introduced LWE problem and completed the hardness reduction from some well-studied worst-case lattice problems quantumly. Then Brakerski et al. proposed MLWE (Definition 5 of Brakerski et al. (2012)) and was further studied in Langlois and Stehlé's research (Langlois and Stehlé 2015). We briefly describe the search/decision versions of the MLWE problem, omitting LWE as it is a special case of MLWE when $n = 1$.

Definition 1 (*MLWE Problem (Adapted from (Brakerski et al. 2012))*) For security parameter λ , let $\ell = \ell(\lambda)$ be an integer dimension, $n = n(\lambda)$ be a power of 2, $f(x) = x^n + 1$ and $q = q(\lambda) \geq 2$ be an integer. Let $\mathcal{R} = \mathbb{Z}[x]/f(x)$ and $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$, $\chi = \chi(\lambda)$ be a distribution over $\mathcal{R}$. For a polynomial vector $s \in \mathcal{R}_q^\ell$, we let $\mathcal{A}_{s,\chi}^{\mathcal{M}}$ denote the distribution over $\mathcal{R}_q^{\ell+1}$ which generates $(a, b = \langle a, s \rangle + e \bmod q)$ by sampling $a \leftarrow \mathcal{R}_q^\ell$ uniformly, $e \leftarrow \chi$. We say $(a, b) \in \mathcal{R}_q^{\ell+1}$ is a $\text{MLWE}_{\ell,n,q,\chi}$ instance.

- The search version of $\text{MLWE}_{\ell,n,q,\chi}$ is: For a fixed vector $s \in \mathcal{R}_q^\ell$, given arbitrarily many samples from $\mathcal{A}_{s,\chi}^{\mathcal{M}}$, to compute s .

- The decision version of $\text{MLWE}_{\ell,n,q,\chi}$ is: Given arbitrarily many samples from $\mathcal{R}_q^{\ell+1}$, to distinguish whether the samples are distributed uniformly from $\mathcal{R}_q^{\ell+1}$ or distributed from $\mathcal{A}_{s,\chi}^{\mathcal{M}}$ for a fixed s .

Subsequently, MLWE has attracted cryptographers' eyes and played a significant role in recent public key cryptographic schemes as its flexible parameters and comparable security. It is popular in the NIST post-quantum cryptography competition (2024), and several finalists in the third round are constructed on MLWE, e.g., the key-encapsulation mechanism CRYSTALS-Kyber (Bos et al. 2018) and the digital signature algorithm CRYSTALS-Dilithium (Bai et al. 2021). We will describe Dilithium in section 2.4.

Small secrets and HNF of η -MLWE

The security and efficiency of an LWE-based cryptosystem depend on its parameter settings. In standard LWE, secret s is drawn randomly from uniform distribution and e is from discrete Gaussian distribution. But for efficiency and lower error growth, some cryptographic schemes require s to be small or even binary. Driven by this requirement, researchers

have explored LWE problems featuring more specialized secret distributions. Applebaum et al. Applebaum et al. (2009) adopt Hermite Normal Form (HNF), which was previously used by Micciancio Micciancio (2001), to obtain an LWE variant where the secret is also from discrete Gaussian distribution. They have proven that such a case where secrets and errors are from the same distribution does not impact the hardness. Moreover, Micciancio and Peikert Micciancio and Peikert (2013) prove that, under the careful choice of parameters, LWE remains hard when the errors are small, even uniformly random from binary distribution (this requires small number of samples). Since then, post-quantum cryptosystems usually employ HNF of LWE with small parameters in practice. Dilithium uses HNF of η -MLWE (It samples secret and noise both from $\mathcal{U}(\mathcal{S}_\eta)$ randomly).¹ Boudgoust Boudgoust (2021) proved the hardness of (non-HNF) η -MLWE when $\eta = 1$ under a proper parameter setting.

Rejection sampling

Rejection sampling is an efficient way to generate valid signatures. For security reasons, the canonical three-move authentication protocol (Abdalla et al. 2002) with specific form requires heavily restricted parameters. Prover has to ensure that the secret key s (bounded by β_1) and response z (bounded by β_2) are both small and $\beta_2 \gg \beta_1$, then the secret recovery and response forgery can be impossible. To achieve this purpose, one should set $\beta_2/\beta_1 > 2^{\omega(\log \lambda)}$ with security parameter λ , this prevents the widespread use of the protocol. Therefore, while designing latticed-based signature schemes from the canonical identification scheme and Fiat-Shamir transformation, Lyubashevsky proposed the rejection sampling technique and developed a general "rejection sampling" lemma (Lyubashevsky 2009, 2012). Rejection sampling relaxes parameter restrictions and ensures that signatures do not expose the private key.

Dilithium furtherly exploited rejection sampling to guarantee security and correctness. For signature value $z = y + cs_1$, rejection condition $\|z\|_\infty \geq \gamma_1 - \beta$ can avoid the dependency of z on the secret key s_1 . For challenge c , rejection conditions $\|r_0\|_\infty \geq \gamma_2 - \beta$, $\|ct_0\|_\infty \geq \gamma_2$ and "number of 1's in h is greater than ω " allow valid signatures to be verified correctly, and also include security considerations. If any condition is not satisfied, signing

¹ η -MLWE means s is from $\mathcal{U}(\mathcal{S}_\eta)$. In $\text{MLWE}_{\ell,n,q,\chi}$, if χ is the uniform distribution over $\mathcal{S}_\eta$, we write it as $\text{MLWE}_{\ell,n,q,\eta}$, and the corresponding distribution is $\mathcal{A}_{s,\eta}^{\mathcal{M}}$. For example, Dilithium uses k samples of HNF of $\text{MLWE}_{\ell,n,q,\eta}$. According to our definitions above, one should know that HNF of η -MLWE and η -MLWE $_{\ell,n,q,\eta}$ are the same thing.

program will abort and restart until one signature value passes all the checks. These checks result in a passing probability of approximately $e^{-256 \cdot \beta(\ell/\gamma_1 + k/\gamma_2)}$ in Sign procedure.² It is around 1/4.25 on average in NIST's security level 2.

Knowledge of dilithium signature

Here we list some properties of Dilithium and show the brief version of Dilithium scheme. As a prominent cryptosystem, Dilithium has a lot of optimization in both theoretical design and implementation, e.g., NTT domain representation and AVX2 instructions, and we omit these details for brevity.

High/low order bits and hints

To reduce the size of public keys and signature values, Dilithium developed the compression technique, which contains “high bits”, “low bits” and “hints”. The goal is that for arbitrary elements $r \in \mathbb{Z}_q$ and another small element $z \in \mathbb{Z}_q$, one can recover the high-order bits of $r + z$ without knowing z . A 1-bit hint plays a key role in this technique. This involves several subroutines containing $\text{Power2Round}_q(r, d)$, $\text{Decompose}_q(r, \alpha)$, $\text{HighBits}_q(r, \alpha)$, $\text{LowBits}_q(r, \alpha)$, $\text{MakeHint}_q(z, r, \alpha)$ and $\text{UseHint}_q(h, r, \alpha)$, also relies on following lemmas. We refer (Bai et al. 2021) for more details and proofs.

Lemma 1 *For positive integers q, α satisfying $q > 2\alpha$, $q \equiv 1 \pmod{\alpha}$ and α even. $\mathbf{r}$ and $\mathbf{z}$ are vectors of elements in $\mathcal{R}_q$ where $\|\mathbf{z}\|_\infty \leq \alpha/2$. Let $\mathbf{h}$ and $\mathbf{h}'$ be vectors of bits. Then algorithm HighBits_q , MakeHint_q , and UseHint_q satisfy the following properties:*

(1) $\text{UseHint}_q(\text{MakeHint}_q(\mathbf{z}, \mathbf{r}, \alpha), \mathbf{r}, \alpha) = \text{HighBits}_q(\mathbf{r} + \mathbf{z}, \alpha)$.

(2) Let $\mathbf{v}_1 = \text{UseHint}_q(\mathbf{h}, \mathbf{r}, \alpha)$. Then $\|\mathbf{r} - \mathbf{v}_1 \cdot \alpha\|_\infty \leq \alpha + 1$. Furthermore, if the number of 1's in $\mathbf{h}$ is ω , then all except at most ω coefficients of $\mathbf{r} - \mathbf{v}_1 \cdot \alpha$ will have magnitude at most $\alpha/2$ after centered reduction modulo q .

(3) For any $\mathbf{h}, \mathbf{h}'$. if $\text{UseHint}_q(\mathbf{h}, \mathbf{r}, \alpha) = \text{UseHint}_q(\mathbf{h}', \mathbf{r}, \alpha)$, then $\mathbf{h} = \mathbf{h}'$.

Lemma 2 *If $\|\mathbf{s}\|_\infty \leq \beta$ and $\|\text{LowBits}_q(\mathbf{r}, \alpha)\|_\infty < \alpha/2 - \beta$, then $\text{HighBits}_q(\mathbf{r}, \alpha) = \text{HighBits}_q(\mathbf{r} + \mathbf{s}, \alpha)$.*

Lemma 3 *Let $(\mathbf{r}_1, \mathbf{r}_0) = \text{Decompose}_q(\mathbf{r}, \alpha)$, $(\mathbf{w}_1, \mathbf{w}_0) = \text{Decompose}_q(\mathbf{r} + \mathbf{s}, \alpha)$, and $\|\mathbf{s}\|_\infty \leq \beta$. Then $\|\mathbf{s} + \mathbf{r}_0\|_\infty < \alpha/2 - \beta \iff \mathbf{w}_1 = \mathbf{r}_1$ and $\|\mathbf{w}_0\|_\infty < \alpha/2 - \beta$.*

Dilithium scheme

In Dilithium, key generation is dependent on k samples of HNF of $\text{MLWE}_{\ell, n, q, \eta}$ and signing procedure relies on MSIS problem. We mainly focus on MLWE problem since our work proposes an improved variant of MLWE, while all the parts related to MSIS remain unchanged. Let q and n be fixed at prime 8380417 and power-of-2 integer 256, respectively. We present a simplified version of Dilithium as follows:

- **Gen:** $(s_1, s_2) \in \mathcal{S}_\eta^\ell \times \mathcal{S}_\eta^k$ and $A \in \mathcal{R}_q^{k \times \ell}$, compute $\mathbf{t} = A s_1 + s_2$, $(\mathbf{t}_1, \mathbf{t}_0) := \text{Power2Round}_q(\mathbf{t}, d)$. Return $\mathbf{pk} = (A, \mathbf{t}_1)$, $\mathbf{sk} = (s_1, s_2, \mathbf{t}_0)$.
- **Sign:** $y \in \tilde{\mathcal{S}}_{\gamma_1}^\ell$, $\mathbf{w} := A\mathbf{y}$ and $\mathbf{w}_1 := \text{HighBits}_q(\mathbf{w}, 2\gamma_2)$. For messages M , compute challenge $c := H(M \| \mathbf{w}_1) \in B_\tau$ using hash function and $\mathbf{z} = \mathbf{y} + c s_1$. Let $\beta = \tau \cdot \eta$, $\mathbf{h} := \text{MakeHint}_q(-c \mathbf{t}_0, \mathbf{w} - c s_2 + c \mathbf{t}_0, 2\gamma_2)$, if $\|\mathbf{z}\|_\infty \geq \gamma_1 - \beta$ or $\|\text{LowBits}_q(\mathbf{w} - c s_2, 2\gamma_2)\|_\infty \geq \gamma_2 - \beta$ or $\|c \mathbf{t}_0\|_\infty \geq \gamma_2$ or the number of 1's in $\mathbf{h}$ is greater than ω , the program aborts and re-runs **sign** procedure. Otherwise return signature value $(\mathbf{z}, \mathbf{h}, c)$.
- **Verify:** To verify a signature value $(\mathbf{z}, \mathbf{h}, c)$ with respect to the message M , one checks $\|\mathbf{z}\|_\infty < \gamma_1 - \beta$ and the number of 1's in $\mathbf{h}$ is less than or equal to ω first, then computes $\mathbf{w}'_1 := \text{UseHint}_q(\mathbf{h}, A\mathbf{z} - c \mathbf{t}_1 \cdot 2^d, 2\gamma_2)$. If the check passed and $c = H(M \| \mathbf{w}'_1)$, return 1, and otherwise return 0.

We refer to (Bai et al. 2021) for correctness, security and more other details.

Our improvements

In this section, we propose a novel technique about Dilithium towards higher efficiency, and obtain an improved signature scheme. Before showing our modified scheme, we first introduce the useful finding: *pre-rejection sampling* technique. The correctness proof is given immediately after the scheme description, and the security discussion is left in the next section.

Pre-rejection sampling

As mentioned above, rejection sampling is one of the primary factors affecting the efficiency of signing procedure in Dilithium (run four times for one valid signature on average). In this subsection, we detail the pre-rejection sampling technique that can reduce the number of rejections. One can view this technique as a strategic

² The expression is just an estimate of the first two rejection conditions. This paper accurately analyzes the rejection probability of the last two conditions for completeness.

approach that removes some rejections from Sign procedure to KeyGen procedure, resulting in increasing the pre-computation time and speeding up the signing time.

The four rejection conditions in Dilithium scheme are distinct. Our pre-rejection sampling is mainly aiming at the third and fourth conditions. For completeness and better understanding, it is necessary to analyze these two rejection conditions first.

Analysis of the third rejection condition

For the third condition $\|ct_0\|_\infty \geq \gamma_2$, we point out that ct_0 is a k -dimensional vector and its entries are all 255-degree polynomials. Each coefficient is the sum of τ independent numbers, and each number is randomly selected from the uniform distribution over the interval $(-2^{d-1}, 2^{d-1}]$. We can compute the probability of this rejection condition, denoted by $\Pr[\|ct_0\|_\infty \geq \gamma_2]$, using the standard Irwin-Hall distribution, as

$$\Pr[\|ct_0\|_\infty \geq \gamma_2] = 1 - \left(\left(\frac{2}{\tau!} \sum_{j=0}^{\lfloor \frac{\gamma_2}{2^d} + \frac{\tau}{2} \rfloor} (-1)^j \binom{\tau}{j} \left(\frac{\gamma_2}{2^d} + \frac{\tau}{2} - j \right)^\tau - 1 \right)^{256 \cdot k} \right). \quad (1)$$

However, the result of Equation (1) is $7.335 \times 10^{-9} \approx 0$ under the parameter set 2 of Dilithium. For parameter sets 3 and 5, this rejection condition is inactive since $\|ct_0\|_\infty \leq \tau \cdot 2^{d-1} < \gamma_2$. In short, the rejecting probability of the third rejection condition is so small under the current parameter settings that it can be ignored.

Analysis of the fourth rejection condition

The fourth condition states that the number of 1's in $\mathbf{h}$ is at most ω . The hint $\mathbf{h}$ is also a k -dimensional vector composed of 255-degree polynomials, whose coefficients are either 0 or 1. A coefficient of 1 indicates that adding $-ct_0$ to $\mathbf{w} - cs_2 + ct_0$ results in carries at those positions, while a coefficient of 0 means that no carry occurs.

Intuitively, the occurrence of carries depends on the sum of $-ct_0$ and $\text{LowBits}_q(\mathbf{w} - cs_2 + ct_0, 2\gamma_2)$. For each position, if the sum is larger than γ_2 , or smaller than or equal to $-\gamma_2$, then carry occurs. In the following, we first estimate the probability of carry occurrence and then calculate the rejection probability for this condition.

The coefficients of $\text{LowBits}_q(\mathbf{w} - cs_2 + ct_0, 2\gamma_2)$ follow a uniform distribution over $(-\gamma_2, \gamma_2]$ because of the randomness. We can treat these coefficients as a uniformly distributed random variable X , and describe its probability density function $f_X(x)$ as $1/(2\gamma_2)$ for $-\gamma_2 < x \leq \gamma_2$ and 0 otherwise.

As previously mentioned, the coefficients of $-ct_0$ follow a (non-standard) Irwin-Hall distribution. We also view the coefficients as random variable Y , whose probability density function is more complicated. Thus, we only represent it abstractly as $f_Y(y)$.

In a similar way, we view the coefficients of the sum of $\text{LowBits}_q(\mathbf{w} - cs_2 + ct_0, 2\gamma_2)$ and $-ct_0$ as random variable Z and denote the probability density function as $f_Z(z)$. Naturally, $Z = X + Y$ has zero expectation and is symmetric, owing to the independence and symmetry of both X and Y . Then the rejection probability of this condition is

$$\begin{aligned} & \Pr[\# \text{ of } 1\text{'s in } \mathbf{h} > \omega] \\ &= 1 - \sum_{j=0}^{\omega} \binom{256 \cdot k}{j} \\ & \quad (2F_Z(-\gamma_2))^j (1 - 2F_Z(-\gamma_2))^{256 \cdot k - j}, \end{aligned}$$

where $F_Z(\cdot)$ is the cumulative distribution function of Z . Unfortunately, computing the exact value of this expression is not straightforward. However, we have experimentally determined that the rejection probability is approximately 1.5% under the parameter set 2 of Dilithium.

To further reduce the rejection probability of the fourth condition, we try to conduct the analysis from the perspective of variance. As X and Y are independent, we have $D(Z) = D(X) + D(Y)$, where $D(\cdot)$ means the variance of the random variables.

The variance of the uniformly random variables X is

$$D(X) = \frac{(\gamma_2 - (-\gamma_2))^2}{12} = \frac{\gamma_2^2}{3}.$$

For Irwin-Hall distributed random variable Y , we have $D(Y) = \sum_{i=1}^{\tau} D(Y_i)$, where Y_i are uniformly distributed random variables over $(-2^{d-1}, 2^{d-1}]$, representing the coefficient distribution of t_0 .³ So we have

$$D(Z) = \frac{\gamma_2^2 + \tau \cdot 4^{d-1}}{3}. \quad (2)$$

Conceptually, the smaller the variance of the random variable Z is, the more concentrated the values are, and the less likely it is that carries will occur when summing $\text{LowBits}_q(\mathbf{w} - cs_2 + ct_0, 2\gamma_2)$ and $-ct_0$. It further means the number of 1's in hint polynomial vector $\mathbf{h}$ tends to be smaller. Therefore, we consider to prescreen t_0 to obtain a distribution with less variance.

More specifically, coefficients of t_0 follow the uniform distribution over the range $(-2^{d-1}, 2^{d-1}]$, which has a

³ Y_i is not exactly the coefficient random variable of t_0 , it may differ by a negative sign. But it doesn't affect the distribution.

Table 1 Parameters of our scheme

Parameter Sets (Security Level)	Param.I (128-bit)	Param.II (192-bit)	Param.III (256-bit)
q : modulus, prime integer	8380417		
n : polynomials degree ($q \equiv 1 \pmod{2n}$)	256		
(k, ℓ) : dimension of $\mathbf{A}$	(4, 4)	(6, 5)	(8, 7)
η : the range of errors and secrets	2	4	2
d : dropped bits of $\mathbf{t}$	13		
τ : number of ± 1 's in $\mathbf{c}$	39	49	60
γ_1 : coefficient range of $\mathbf{y}$	2^{17}	2^{19}	2^{19}
γ_2 : low-order rounding range	$(q-1)/88$	$(q-1)/32$	$(q-1)/32$
ω : maximum of the number of 1's in the hint $\mathbf{h}$	80	55	75
β : upper bound of the ℓ_1 -norm of $\mathbf{cs}_1$ and $\mathbf{cs}_2$ ($\beta = \tau\eta$)	78	196	120
σ : pre-rejection parameter ($3\sigma < 2^{d-1}$)	$2^{d-1}/3.225$	$2^{d-1}/3.188$	$2^{d-1}/3.180$

very large variance. We observe that if we use another distribution with a thin tail instead, the variance would be smaller and the coefficients would be more concentrated.

Pre-rejection sampling

In order to force the coefficients of $\mathbf{t}_0$ into a thin-tailed and concentrated distribution, denoted by $\mathcal{D}$, that is close to the uniform distribution,⁴ we use the empirical rule (a.k.a. 3σ rule (Pukelsheim 1994, 2024)) to pre-select $\mathbf{t}_0$ in KeyGen procedure. As a rule of thumb, 3σ rule is widely used in practice, especially in the case of normal distribution. Even though $\mathcal{D}$ is not a normal distribution, we

reasonably state that it can be a symmetrically unimodal distribution, which is enough to say that the vast majority of the values lie within three standard deviations of the mean.

As a priori assumption, we assume that for a random variable from distribution $\mathcal{D}$, there should be at least 95% probability that it falls into $[-3\sigma, 3\sigma]$. To be specific, in KeyGen stage, we check all $256 \cdot k$ coefficients of $\mathbf{t}_0$: For a k -dimensional polynomial vector $\mathbf{t}_0$, if the number of coefficients outside 3σ -region is more than 5%, we think this $\mathbf{t}_0$ is not follow distribution $\mathcal{D}$, and restart the

KeyGen procedure to generate new $\mathbf{t}_0$, until the checking passes. This sub-routine of checking $\mathbf{t}_0$ is summarized in Algorithm 1.

Algorithm 1 CheckT($\mathbf{t}_0, \sigma$): check whether $\mathbf{t}_0$ satisfies distribution $\mathcal{D}$

Require: $\mathbf{t}_0 \in \mathcal{R}^k$, $3\sigma < 2^{d-1}$ $\triangleright \sigma$ is the standard deviation of $\mathcal{D}$
Ensure: $\{0, 1\}$ $\triangleright$ false, true

```

1: counter = 0
2: for  $i = 0, \dots, k-1$  do
3:   for  $j = 0, \dots, 255$  do
4:     if  $\mathbf{t}_0[i].coef[j] < -3\sigma$  or  $\mathbf{t}_0[i].coef[j] > 3\sigma$  then
5:       counter++
6:     end if
7:   end for
8: end for
9: if counter  $> 256 \cdot k \cdot 5\%$  then
10:  return 0
11: end if
12: return 1

```

⁴ Note that normal distribution is infeasible. It is somewhat far from the uniform distribution, which makes the pre-rejection sampling difficult to implement efficiently.

After pre-rejection sampling, we view the coefficients of $\mathbf{t}_0$ as random variables that follow distribution $\mathcal{D}$ with standard deviation σ , then the variance of random variable Z is

$$D(Z) = \frac{\gamma_2^2}{3} + \tau\sigma^2,$$

which is significantly smaller than Equation (2) since we require $3\sigma < 2^{d-1}$.

Signature scheme

Applying pre-rejection sampling to KeyGen step, we get our modified signature scheme (see Signature Scheme). Our scheme is parameterized by $n, q, \ell, k, \eta, d, \tau, \gamma_1, \gamma_2, \omega, \beta (= \tau\eta)$ and σ (we keep σ in secret), see Table 1 for a detailed description of all system parameters. We focus on describing the key generation stage because other parts are the same as Dilithium.

Scheme description

Here we present our signature scheme (mainly KeyGen algorithm). First, one generates secret $\mathbf{s}_1$, error $\mathbf{s}_2$, and matrix $\mathbf{A}$ from random seeds via hash functions (see (Bai et al. 2021) for the functions ExpandS and ExpandA). Then compute $\mathbf{t}$ according to MLWE $_{\ell, n, q, \eta}$. To compress the public key, $\mathbf{t}$ is separated into high-bit part $\mathbf{t}_1$ and low-bit part $\mathbf{t}_0$. The subroutine CheckT implements pre-rejection sampling, which is beneficial to increase the probability of passing the rejection condition in Sign procedure. We emphasize that CheckT has not impacted the security of Sign procedure in practice because we conceal σ and $\mathbf{t}_0$ in secret.

Signature Scheme

KeyGen:

```

01:  $\zeta \leftarrow \{0, 1\}^{256}, \kappa = 0$ 
02:  $(\rho, \varsigma, K) \in \{0, 1\}^{256+512+256} := \text{H}(\zeta)$ 
03:  $\mathbf{s}_1 \in \mathcal{S}_{\eta}^{\ell} := \text{ExpandS}(\varsigma, \kappa)$ 
04:  $\mathbf{s}_2 \in \mathcal{S}_{\eta}^k := \text{ExpandS}(\varsigma, \kappa + \ell)$ 
05:  $\mathbf{A} \in \mathcal{R}_{\eta}^{k \times \ell} := \text{ExpandA}(\rho + \kappa)$ 
06:  $\mathbf{t} := \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$ 
07:  $(\mathbf{t}_1, \mathbf{t}_0) := \text{Power2Round}_q(\mathbf{t}, d)$ 
08: If  $\text{CheckT}(\mathbf{t}_0, \sigma) = 0$ :  $\kappa = \kappa + \ell + k$ , goto step 03
09:  $tr \in \{0, 1\}^{256} := \text{H}(\rho \parallel \mathbf{t}_1)$ 
10:  $(pk = (\rho, \mathbf{t}_1), sk = (K, tr, \mathbf{s}_1, \mathbf{s}_2, \mathbf{t}_0, \sigma))$ 

```

Sign and Verify is same as Bai et al. (2021) so we omit here.

We also emphasize that our scheme is compatible with Dilithium, and transforming Dilithium to our scheme is so easy and no loss. Adding a CheckT procedure to KeyGen, without affecting other parts. One

can use the key pairs generated by our KeyGen procedure, but for the Sign and Verify procedures in the Dilithium signature systems. Vice versa, the public keys and signatures of Dilithium can also be used as the inputs of our system. Even for an on-going signature system, the issued Dilithium certificate can still be correctly verified by our new scheme.

Correctness of our scheme

For a signature value $(\mathbf{z}, \mathbf{h}, \tilde{c})$ produced by Sign procedure, we say it can pass the Verify procedure. According to Lemma 1, we know that

$$\text{UseHint}_q(\mathbf{h}, \mathbf{w} - \mathbf{cs}_2 + \mathbf{ct}_0, 2\gamma_2) = \text{HighBits}_q(\mathbf{w} - \mathbf{cs}_2, 2\gamma_2).$$

We also have that

$$\mathbf{w} - \mathbf{cs}_2 = \mathbf{A}\mathbf{y} - \mathbf{cs}_2 = \mathbf{A}(\mathbf{z} - \mathbf{cs}_1) - \mathbf{cs}_2 = \mathbf{A}\mathbf{z} - \mathbf{ct}$$

since $\mathbf{w} = \mathbf{A}\mathbf{y}$, $\mathbf{z} = \mathbf{y} + \mathbf{cs}_1$ and $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$, so $\mathbf{w} - \mathbf{cs}_2 + \mathbf{ct}_0 = \mathbf{A}\mathbf{z} - \mathbf{ct}_1 \cdot 2^d$. Therefore the verifier computes

$$\text{UseHint}_q(\mathbf{h}, \mathbf{A}\mathbf{z} - \mathbf{ct}_1 \cdot 2^d, 2\gamma_2) = \text{HighBits}_q(\mathbf{w} - \mathbf{cs}_2, 2\gamma_2).$$

Furthermore, we know $\text{LowBits}_q(\mathbf{w} - \mathbf{cs}_2, 2\gamma_2) < \gamma_2 - \beta$ in Sign and $\|\mathbf{cs}_2\|_{\infty} \leq \beta$ because of $\beta = \tau\eta$, then Lemma 2 implies that

$$\begin{aligned}
& \text{HighBits}_q(\mathbf{w} - \mathbf{cs}_2, 2\gamma_2) \\
&= \text{HighBits}_q(\mathbf{w} - \mathbf{cs}_2 + \mathbf{cs}_2, 2\gamma_2) \\
&= \text{HighBits}_q(\mathbf{w}, 2\gamma_2) \\
&= \mathbf{w}_1.
\end{aligned}$$

Consequently, in Verify procedure, $\mathbf{w}'_1$ is equal to $\mathbf{w}_1$ of Sign procedure, and then the challenge value c equals to the output of hash function, which means the signature value would be accepted.

Security

In this section, we analyze the security of our signature scheme. We first discuss the c -selected MLWE problem in subsection 4.1. Then in subsection 4.2, we complete the reduction of our scheme from Dilithium using the hardness of selected MLWE problem.

c-Selected MLWE

In our KeyGen procedure, we use pre-rejection sampling technique to enforce the coefficients of $\mathbf{t}_0$ to follow a distribution $\mathcal{D}$ instead of the uniform distribution. Such public key $(\mathbf{A}, \mathbf{t})$ is not a standard MLWE instance anymore. The possible set of $\mathbf{t}_0$ (also $\mathbf{t}$) is smaller. Our goal is

to prove that this type of selected MLWE instance $(A, \mathbf{t})$ is also secure.

For the sake of description, we introduce *c-select function* first.

Definition 2 (*c-Select Function*) Let $0 < c \leq 1$ be a constant, S be a finite set and $S' \subseteq S$. For a computable function f , we say it is a *c-select function* of S if

$$f : S \rightarrow S' \text{ and } \frac{|S'|}{|S|} = c,$$

where $|S|$ represents the size of set S . We also say S' is the *c-selected set* of S if such relations satisfied, and write S' as $f(S)$.

As described above, public key $\mathbf{t}$ of MLWE is uniformly distributed from $\mathcal{R}_q$ according to the MLWE assumption and we select it by function **CheckT**. Next, we construct *c-selected MLWE problem* using this terminology.

Definition 3 (*c-Selected MLWE Problem*) For security parameter λ , let $\ell = \ell(\lambda)$ be an integer dimension, $n = n(\lambda)$ be a power of 2, $f(x) = x^n + 1$ and $q = q(\lambda) \geq 2$ be an integer. Let $\mathcal{R} = \mathbb{Z}[x]/f(x)$ and $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$, $\chi = \chi(\lambda)$ be a distribution over $\mathcal{R}$. Let g be a *c-select function* of $\mathcal{R}_q^{\ell+1}$ for a positive constant $c \leq 1$. For a polynomial vector $\mathbf{s} \in \mathcal{R}_q^\ell$, we let $\mathcal{A}_{s,\chi,g}^M$ denote the distribution over $g(\mathcal{R}_q^{\ell+1})$ which generates $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q)$ by sampling $\mathbf{a} \leftarrow \mathcal{R}_q^\ell$

uniformly, $e \leftarrow \chi$. We say such a pair $(\mathbf{a}, b) \in g(\mathcal{R}_q^{\ell+1})$ is a *c-selected MLWE* $_{\ell,n,q,\chi}$ instance.

- The search version of *c-selected MLWE* $_{\ell,n,q,\chi}$ is: For a fixed vector $\mathbf{s} \in \mathcal{R}_q^\ell$, given arbitrarily many samples from $\mathcal{A}_{s,\chi,g}^M$, to compute $\mathbf{s}$.

- The decision version of *c-selected MLWE* $_{\ell,n,q,\chi}$ is: Given arbitrarily many samples from $g(\mathcal{R}_q^{\ell+1})$, to distinguish whether the samples are distributed uniformly from $g(\mathcal{R}_q^{\ell+1})$

or distributed from $\mathcal{A}_{s,\chi,g}^M$ for a fixed $\mathbf{s}$.

To illustrate the hardness of *c-selected MLWE problem*, we conduct a security reduction from MLWE assumption. The main result is presented in Theorem 1. Note that the hardness reduction from decision version to search version is trivial, so we only consider the decision version in our proof.

Theorem 1 Let ℓ, n, q, χ are integers defined as in Definition 3 and $c \leq 1$ be a positive constant. g is a *c-select*

function of $\mathcal{R}_q^{\ell+1}$. There exist a probabilistic polynomial-time reduction from (decision) *MLWE* $_{\ell,n,q,\chi}$ to (decision) *c-selected MLWE* $_{\ell,n,q,\chi}$.

Proof In order to prove the theorem, we define several hybrid distributions:

- H_0 : $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q) \in g(\mathcal{R}_q^{\ell+1})$, as defined in *c-selected MLWE* $_{\ell,n,q,\chi}$.
- H_1 : $(\mathbf{a}, b) \leftarrow \mathcal{U}(g(\mathcal{R}_q^{\ell+1}))$.
- J_0 : $(\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + e \bmod q) \in \mathcal{R}_q^{\ell+1}$, as defined in *MLWE* $_{\ell,n,q,\chi}$.
- J_1 : $(\mathbf{a}, b) \leftarrow \mathcal{U}(\mathcal{R}_q^{\ell+1})$.

The overall idea is to show that if J_0 and J_1 are indistinguishable, then H_0 and H_1 are indistinguishable as well.

For decisional *MLWE* $_{\ell,n,q,\chi}$, we say that the advantage of any probabilistic polynomial-time (PPT) attacker A in solving the problem (distinguishing it from uniform distribution) is

$$\text{Adv}_{\ell,n,q,\chi}^{\text{MLWE}}(A) := \left| 2\Pr[A(J_1) = 1] - 1 \right|, \quad (3)$$

and by the hardness assumption of *MLWE* $_{\ell,n,q,\chi}$, we have

$$\text{Adv}_{\ell,n,q,\chi}^{\text{MLWE}}(A) \leq \text{negl}(\lambda) \quad (4)$$

where λ is the security parameter.

On the other hand, for any PPT adversary B who attempts to distinguish the *c-selected MLWE* $_{\ell,n,q,\chi}$ and the uniform distribution over selected set (i.e. distribution H_1), we consider to instantiate the distinguishing algorithm of adversaries A by employing B as follow:

Adversary A

Input: J_0, J_1 and an instance $\mathbf{c} = (\mathbf{a}, b) \in \mathcal{R}_q^{\ell+1}$.

Output: 0 (means J_0) or 1 (means J_1).

1: Check $\mathbf{c}$. If $\mathbf{c} \notin g(\mathcal{R}_q^{\ell+1})$, $r \leftarrow \mathcal{U}(\{0, 1\})$ and return r .

2: Run B . If $B(\mathbf{c}) = 0$, return 0. If $B(\mathbf{c}) = 1$, return 1.

3: Else return $\perp$.

According to the construction of adversary A , we know that for input $\mathbf{c} \in J_0$, A returns 0 with probability $c \cdot \Pr[B(H_0) = 0] + \frac{1}{2} \cdot (1 - c)$ since H_0 is the *c-selected set* of J_0 . Similarly, for input $\mathbf{c} \in J_1$, A returns 1 with probability $c \cdot \Pr[B(H_1) = 1] + \frac{1}{2} \cdot (1 - c)$.

We say the advantage for B to distinguish H_0 and H_1 is

$$\begin{aligned}
\text{Adv}_{\ell,n,q,\chi}^{c\text{-MLWE}}(\mathbf{B}) &:= \left| 2\Pr\left[\mathbf{B}(\mathbf{H}_r) = r \mid r \xleftarrow{\$}\{0,1\}\right] - 1 \right| \\
&= \left| 2\left(\frac{1}{2}\Pr[\mathbf{B}(\mathbf{H}_0) = 0 \mid r = 0] + \frac{1}{2}\Pr[\mathbf{B}(\mathbf{H}_1) = 1 \mid r = 1]\right) - 1 \right| \\
&= |\Pr[\mathbf{B}(\mathbf{H}_0) = 0] + \Pr[\mathbf{B}(\mathbf{H}_1) = 1] - 1|.
\end{aligned} \tag{5}$$

Using equations (3), (4) and (5), and the construction of adversary $\mathbb{A}$, we have

$$\begin{aligned}
\text{negl}(\lambda) &\geq \left| 2\Pr\left[\mathbf{A}(\mathbf{J}_r) = r \mid r \xleftarrow{\$}\{0,1\}\right] - 1 \right| \\
&= |\Pr[\mathbf{A}(\mathbf{J}_0) = 0] + \Pr[\mathbf{A}(\mathbf{J}_1) = 1] - 1| \\
&= \left| c \cdot \Pr[\mathbf{B}(\mathbf{H}_0) = 0] + \frac{1}{2} \cdot (1 - c) + c \cdot \Pr[\mathbf{B}(\mathbf{H}_1) = 1] + \frac{1}{2} \cdot (1 - c) - 1 \right| \\
&= c \cdot |\Pr[\mathbf{B}(\mathbf{H}_0) = 0] + \Pr[\mathbf{B}(\mathbf{H}_1) = 1] - 1| \\
&= c \cdot \text{Adv}_{\ell,n,q,\chi}^{c\text{-MLWE}}(\mathbf{B}).
\end{aligned}$$

Note that c is a constant, so $\text{Adv}_{\ell,n,q,\chi}^{c\text{-MLWE}}(\mathbf{B}) \leq \text{negl}(\lambda)$. $\square$

With the hardness assumption of the c -selected MLWE, we have the follow lemma that implies our public key is indistinguishable from the instance of uniform distribution over c -selected set.

Lemma 4 *The pre-rejection sampling algorithm CheckT in our KeyGen procedure is a c -selected function of $\eta\text{-MLWE}_{\ell,n,q,\eta}$ for a constant c .*

Proof The proof of lemma 4 is obvious according to the Definition 2 except for the constant c . We point out that the upper bound of running time of the select function CheckT must be a constant, otherwise, the efficiency of KeyGen is unacceptable. This ensures the existence of constant c . $\square$

Remark 1 Dilithium employs the $\eta\text{-MLWE}_{\ell,n,q,\eta}$ but we only provide the reduction of c -selected MLWE from the standard MLWE problem in Theorem 1 for generality. It is directly for one to complete a similar security proof from $\eta\text{-MLWE}_{\ell,n,q,\eta}$ if necessary.

Security reduction of signature scheme

The security of our signature scheme is based on Dilithium, where the latter was analyzed in Kiltz et al. (2018). It was shown that Dilithium is SUF-CMA (Strong UnForgeability under Chosen Message Attacks) in

QROM. We claim that our signature scheme is also SUF-CMA in QROM, and analyze the security in both practice and theory.

Note that our scheme differs from Dilithium only due

to the part of the secret key $\mathbf{t}_0$, which is unknowable for the adversary (he only gets $\mathbf{A}$ and $\mathbf{t}_1$), so the adversary can not make more effective attacks in the real scheme. In the security reduction of Dilithium scheme, however, it is assumed that the adversary gets $\mathbf{t} = \mathbf{A}\mathbf{s}_1 + \mathbf{s}_2$ (i.e. both $\mathbf{t}_0$ and $\mathbf{t}_1$) for easy proof and even more secure concern. Therefore, we also need to analyze what happens to the security in the case of exposing $\mathbf{t}_0$. Theorem 2 ensures the SUF-CMA of our scheme in QROM.

Theorem 2 *If $\mathbb{H}$ is a quantum-accessible perfect hash function, the advantage of any adversary $\mathbb{A}$ breaking the SUF-CMA security of our signature scheme is as small as the advantage of adversaries breaking Dilithium scheme.*

Proof As mentioned above, the only difference between our signature scheme and Dilithium is pre-rejection sampling in KeyGen, which enforces $\mathbf{t}_0$ possible smaller. So the hardness of secret recovery relying on c -selected MLWE problem instead of HNF of MLWE problem. Notice that this is only a pre-selection in key generation without affecting signing and verification, and the signature value $\mathbf{z}$, challenge value c and hint $\mathbf{h}$ are still zero-knowledge.

Therefore the security proof of our scheme is fairly consistent with that of Dilithium in Kiltz et al. (2018) and Bai et al. (2021). Intuitively, the c -selected MLWE

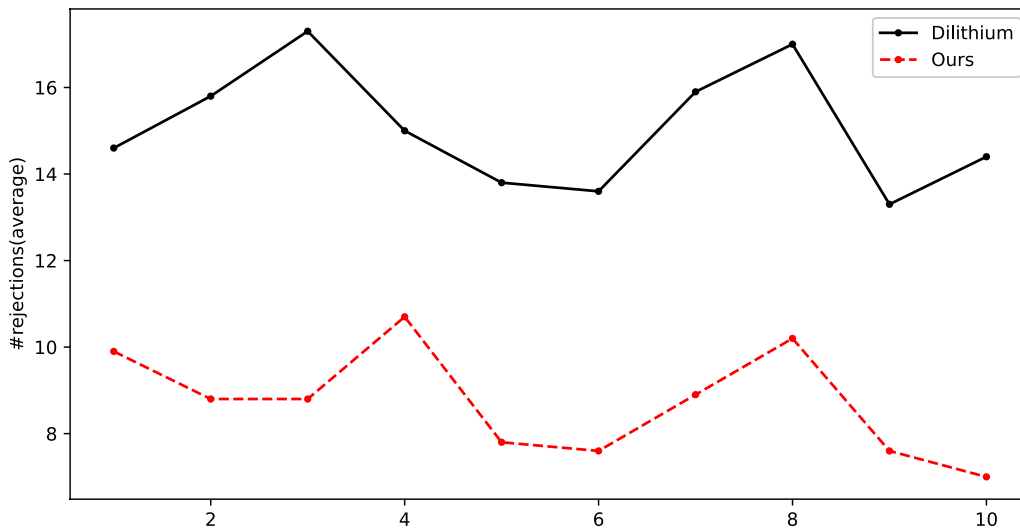


Fig. 1 Average number of rejections in terms of the fourth condition before generating 1000 valid signatures (The horizontal axis represents 10 rounds of experiments). Each value in this figure is an average of 10 repetitions

assumption is required to protect against key-recovery, the SelfTargetMSIS assumption (defined in Kiltz et al. (2018)) ensures that message forgery is impossible and the MSIS assumption is needed for strong unforgeability.

In QROM, H is a quantum random oracle. Let the advantage of an adversary A breaking the SUF-CMA security of our scheme be $\text{Adv}_{\text{Ours}}^{\text{SUF-CMA}}(A)$, we have that the advantage of adversary A breaking the SUF-CMA security of Dilithium signature is $\text{Adv}_{\text{Dilithium}}^{\text{SUF-CMA}}(A)$. It is known that⁵

$$\begin{aligned} \text{Adv}_{\text{Dilithium}}^{\text{SUF-CMA}}(A) &\leq \text{Adv}_{k,\ell,D}^{\text{MLWE}}(E) \\ &+ \text{Adv}_{H,k,\ell+1,\zeta}^{\text{SelfTargetMSIS}}(C) + \text{Adv}_{k,\ell,\zeta'}^{\text{MSIS}}(D) + 2^{-254} \end{aligned}$$

for D a uniform distribution over S_η , and $\zeta = \max\{\gamma_1 - \beta, 2\gamma_2 + 1 + 2^{d-1} \cdot \tau\}$, $\zeta' = \max\{2(\gamma_1 - \beta), 4\gamma_2 + 2\}$.

Recall the security proof of Dilithium (see section 4 of Kiltz et al. (2018)), we need to modify the lossy key generator by adding the `CheckT` step (as showed in our `KeyGen`), then the lossiness property is depended on the hardness of c -selected MLWE assumption. Keeping other part identical, one can obtain the SUF-CMA of our scheme. As a result we have

$$\begin{aligned} \text{Adv}_{\text{Ours}}^{\text{SUF-CMA}}(A) &\leq \text{Adv}_{k,\ell,D}^{c\text{-MLWE}}(B) \\ &+ \text{Adv}_{H,k,\ell+1,\zeta}^{\text{SelfTargetMSIS}}(C) + \text{Adv}_{k,\ell,\zeta'}^{\text{MSIS}}(D) + 2^{-254}, \end{aligned}$$

⁵ We omit parameter n and q in this inequality as Dilithium fixes them invariant.

Theorem 1 and Lemma 4 imply that c -selected MLWE is as hard as MLWE problem, and therefore the SUF-CMA security of our signature scheme is as hard as Dilithium scheme. $\square$

Parameter selection and implementation

We show our parameters and implementation results in this section. The reference parameters of our signature system are summarized in Table 1. We implement our signature system and compare it with Dilithium, the results show that our pre-rejection sampling exactly excludes some bad t_0 's and reduces the number of rejections, thus causing the decrement of running time of the Sign procedure.

Parameter selection

In Table 1, we present three parameter sets corresponding to NIST's different security levels (Parameter set I, II and III correspond to NIST security levels 2,3 and 5, respectively). For a fair comparison, parameter values in these three parameter sets also correspond to that of Dilithium. The parameters σ is introduced from the pre-rejection sampling technique, whereas other parameters are the same as Dilithium. We conduct all the experiments and obtain the results of this paper under the parameters presented in Table 1.

Implementation performance

We implement our signature scheme through the C program with NTT technique and AVX2 instructions. Concretely, we first refer to the publicly available

Table 2 Performance comparison. Each value is the average of 10,000 times runs

Average Cycles (cycles/ticks)	Dilithium			Our Scheme		
	KeyGen	Sign	Verify	KeyGen (Efficiency Loss)	Sign (Speed-up)	Verify
Unoptimized Reference Code (without C compiler optimization)						
Param. I	546,011	2,575,885	625,823	45,200,423 (82.8×)	2,548,841 (1.06%)	619,168
Param. II	951,087	4,208,175	1,007,022	6,761,053 (7.1×)	4,199,728 (0.20%)	1,001,460
Param.III	1,511,992	5,086,526	1,645,333	6,458,606 (4.3×)	5,056,831 (0.59%)	1,639,657
AVX2						
Param. I	49,405	148,168	52,973	2,242,019 (45.4×)	146,684 (1.01%)	51,958
Param. II	84,838	236,530	86,709	355,056 (4.19×)	236,152 (0.16%)	85,693
Param.III	134,153	285,708	134,906	372,374 (2.8×)	285,140 (0.20%)	133,675
AVX2+AES						
Param. I	29,065	111,509	36,022	1,521,556 (52.4×)	110,276 (1.12%)	36,222
Param. II	46,653	163,578	52,528	238,773 (5.1×)	163,170 (0.25%)	52,815
Param.III	69,129	193,840	77,288	223,140 (3.2×)	193,288 (0.29%)	76,723

Dilithium library (Dilithium 2022), which contains the official reference implementation of the Dilithium signature scheme (the current implementation is Version 3.1 of Dilithium (Bai et al. 2021)). Then we implement our scheme based on this repository. The main code development is the adjustment of the key generation part. In this section, the experimental results of Dilithium were produced by the repository (Dilithium 2022) while the results of the proposed scheme were from our modified implementation, which is available at (2024). All the experiments are equipped with Intel(R) Xeon(R) Gold 6230R 2.10GHz CPU and 252GB memory, running Ubuntu 22.04.1. The compiler we used is gcc 11.4.0. Experimental results show that, compared with the Dilithium signature scheme, our proposed scheme achieves faster signing speed at the cost of increased key generation time, just as our theoretical description above.

Firstly, we show the reduction of the rejection probability of the fourth rejection condition. To capture the small rejection probability, we run the two schemes 1,000 times respectively, and count the total number of rejections with respect to the fourth rejection condition in Sign stage. Moreover, to minimize the impact of random errors, each result value is the average of 10 repetitions. Figure 1 shows the results of 10 rounds

experiments for Param. I. The number of rejections with respect to the fourth condition in Sign procedure of our scheme is around 8 while that of Dilithium is around 15, which means that we reduce the rejection probability of the fourth condition from 1.5% to 0.8%. For Param. II and Param.III, the rejection probabilities of the fourth condition in our proposed scheme are 0.28% and 0.64% respectively, while those of Dilithium are 0.36% and 0.77% respectively.

Next, we discuss the implementation effect of the overall scheme. Table 2 shows the average cycles of each stage with optimizations in different levels. We run each subroutine 10,000 times to weaken the randomness of running time. We mainly focus on the the results of the Sign and KeyGen stages in Table 2 since we speed up the signing by sacrificing the pre-computation time of key generation and keep the verification procedure unchanged.

For Param. I, we achieve around 1% signing speed-up with an 82.8× KeyGen time in unoptimized reference code. In AVX2 and AVX2+AES mode, it achieves 45.4× and 52.4× KeyGen time respectively, also with around 1% signing speed-up. This efficiency loss of up to 82.8× in KeyGen is supportable because one key generation of the original Dilithium scheme only requires less than 0.5

milliseconds, and once a key is generated, it can be used in many times of signings and verifications. Therefore, the speed-up of signing procedure is more important. All the mentioned important results are highlighted in Table 2 by bold.

For other two parameter sets, the efficiency improvement of signing is relatively small, because the fourth rejection condition accounts for a small proportion of the total rejection probability under these parameter settings. Therefore, we set larger values of σ in these two parameter sets for a lower efficiency loss in key generation.

Another straightforward way to achieve faster signing speed is lowering the value of σ . However, probability of passing the pre-rejections is sensitive with respect to σ and small value makes KeyGen difficult. Our choice of parameters in Table 1 is relatively reasonable. It has an acceptable running-time for KeyGen stage, and a slightly smaller σ will drastically increase the time of KeyGen and have almost little gains for signing.

Other related works

In recent years, there have been many explorations in lattice-based signatures, aiming to find a balance between security and practicality, as well as a balance between the sizes of public keys and signatures, and computation time. For an instance, Cheon et al. Cheon et al. (2023, 2023) propose a new lattice-based signature scheme called HAETAE. HAETAE uses a bimodal distribution for the rejection sampling, achieving up to 39% smaller signature and key sizes than Dilithium, albeit at the cost of an approximately 500% increase in sign generation time. This reduction in signature size allows it to be transmitted in a single UDP or TCP datagram, thereby enhancing practicality.

Other lattice-based signature schemes in recent years also exhibit similar trade-offs to HAETAE. Schemes with larger signature and public key sizes than Dilithium include EagleSign Sow et al. (2023), Raccoon Pino et al. (2023), SQUIRRELS Espitau et al. (2023), and HUFU Yu et al. (2023). Some signature schemes optimize signature and key sizes but compromise signature computation time, e.g., another NIST-selected digital signature algorithm Falcon Fouque et al. (2022). Some proposed algorithms like EHTv4 Semaev and Feussner (2023), Mitaka Espitau et al. (2022) and EagleSign Sow et al. (2023) are threatened by reported attacks Pulles and Tibouchi (2024); Ryan and Suhl (2023); Prest (2023). For a more detailed comparison of these recent algorithms, readers can refer to the Appendix A of paper (Cheon et al. 2023), or the NIST Standardization of Additional Digital Signature Schemes (2024). Consequently, we believe that Dilithium still represent (at least

one of) current state-of-the-art lattice-based signature scheme at the time of this writing.

Conclusion and further work

In this work, we obtain a Fiat-Shamir signature scheme based on Dilithium using a novel idea, pre-rejection sampling, which improves signing efficiency without any security loss. More precisely, pre-rejection sampling reduces the number of rejections in the signing stage, and the new hardness assumption c -selected MLWE provides security support. So our signature system has a better signing efficiency than Dilithium, at the expense of slower KeyGen. Pre-rejection sampling increases the running time of KeyGen, but in most real systems, the key will be used to sign many times, so the increased cost in KeyGen time can be amortized evenly. We also emphasize that our scheme is totally compatible with Dilithium, one can use the key pairs generated by our KeyGen procedure, but for the Sign and Verify procedures in the Dilithium signature systems.

An important but troublesome problem is increasing the passing probability of the first-round rejection conditions, i.e., the first and second conditions. Dilithium (Bai et al. 2021) gives a rough estimate by assuming coefficients of cs_1 and cs_2 are uniformly distributed, which is enough to compute the rejection probability. Nevertheless, thinking about it from other perspectives may yield interesting results. Another issue is that the security of c -selected MLWE is a theoretical analysis, evaluating the concrete security level of proposed c -selected MLWE problem using all the popular lattice reduction attack algorithms still remains to be done. We will dedicate to solving these in the following work.

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Author contributions

LY proposed the pre-rejection sampling technique. LY and ML completed the experiments and manuscript together. MW participated in problem discussions and improvements of the manuscript. All authors read and approved the final manuscript.

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Availability of data and materials

Not applicable.

Declarations

Competing interests

None of the authors have any competing interests in the manuscript.

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